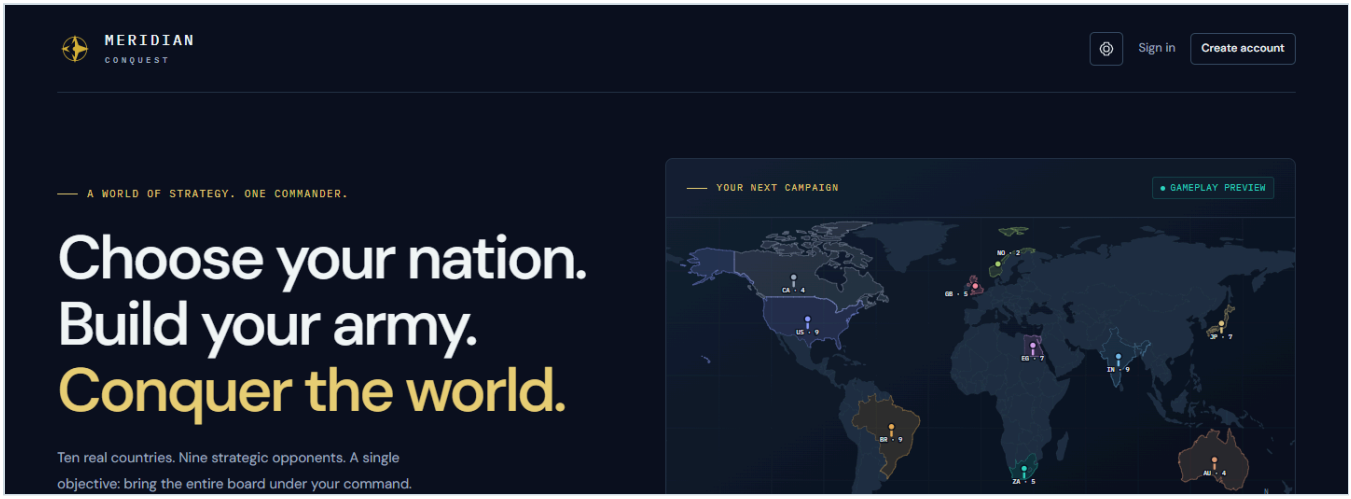
Meridian Conquest

Project documentation · INF4001N entrance assignment

A turn-based strategy game in which ten real countries compete until one commander controls the entire board.

Live application: https://meridian-conquest.vercel.app
Repository: https://github.com/Muhammed112/Meridian_Conquest_UI



Overview

The project implements the University of Cape Town INF4001N geopolitics case study. A player selects a country, receives a population-based army and faces nine strategic NPC commanders. Each turn combines reinforcement decisions, optional transfers and one attack target. The victory condition is ownership of all ten countries.

Extra features include multiplayer rooms, temporary NPC alliances, an interactive adviser tutorial, replay, sound and animation, and public live viewing. Gemini provides commentary about recorded events without deciding combat outcomes.

Contents

1. Overview and project information
2. System architecture and rationale
3. Key features and game rules
4. Getting started and environment configuration
5. System components and database design
6. User roles and workflows
7. Verification, troubleshooting and limitations
8. Project structure, documentation and acknowledgements

Prepared 5 November 2026 · Application release c10feed · Author: Muhammed (Muhammed112)

System architecture

Browser game logic with managed Firebase services

The application uses HTML, CSS and native JavaScript ES modules. Vercel serves the maintained `dist` source directly. There is no frontend compilation step or deployed custom game server.

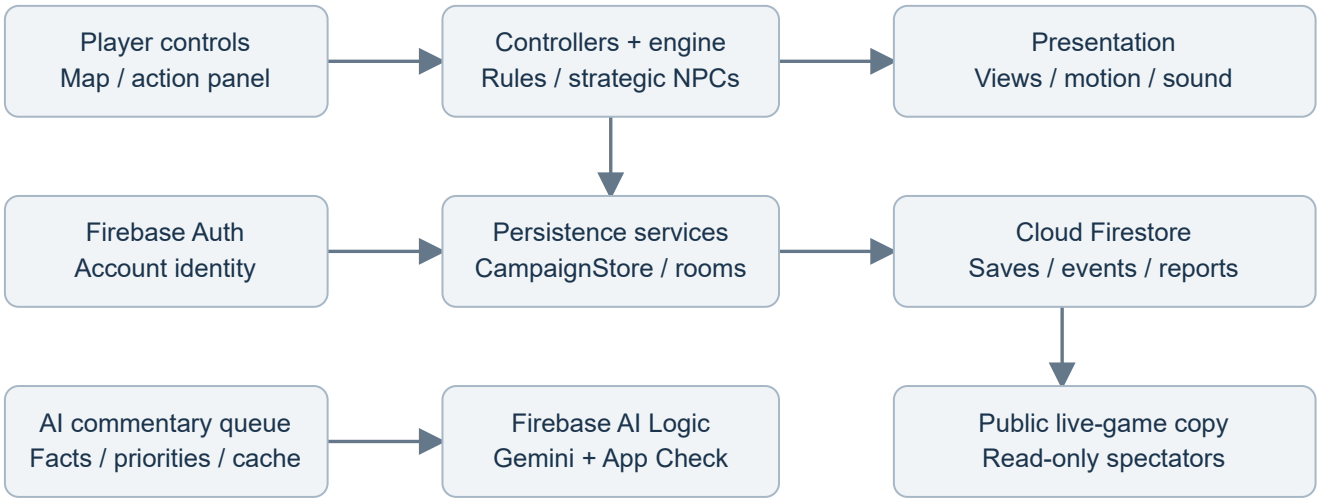


Figure 1. Main responsibilities and external services. Game state and AI narration follow separate paths.

Architecture rationale

Decision	Reason
Separate game engine	Rules can be tested without a browser, database or AI model.
Shared views and controllers	Single-player and multiplayer reuse controls while resolving the current viewer’s permissions.
Cloud Firestore	Document saves, append-only events and live listeners suit independent campaigns and shared rooms.
Firebase Authentication	Credentials remain with the authentication service; application passwords are not stored in Firestore.
Independent commentary queue	Slow or unavailable model requests do not block game turns.
Bundled assets and snapshots	The map, flags and country data remain available when a source-data refresh fails.

Event and narration flow

1. A controller receives a player action or an NPC plan and calls the game engine.
2. The engine checks legal moves, changes the board and records factual events.
3. Views display the board; presentation timing hides casualties until the dice settle.
4. The persistence service saves the board and immutable facts under the permitted account or room.
5. Eligible events enter the priority queue. Gemini receives compact event-time facts through Firebase AI Logic.
6. Accepted narration is stored separately and shown in the AI commentary tab. An explicit public copy can deliver available reports to spectators.

Key features and game rules

Required gameplay and extensions

Core functionality

- Ten distinct countries: the player chooses one and nine rivals are drawn randomly. Multiple campaigns can be saved independently.
- Starting armies follow the brief's population bands. The catalogue contains 233 selectable countries and territories.
- All commanders roll for the first turn. Tied leaders reroll; the final turn-order preview appears before play starts.
- NPCs plan reinforcements, transfers, attacks and retreats. Their reasons remain visible during fast-forward.
- Paired dice, casualties, captures, elimination and victory are displayed during play.

Rule	Implementation
Starting soldiers	1–9 soldiers using population upper bounds of 5, 10, 25, 50, 75, 100, 125 and 150 million; larger populations receive 9.
Reinforcements	From round two: 2 soldiers per owned country, allocated among the commander's holdings.
Attacks	One enemy target per turn. Any location is reachable. The attacking country needs at least 2 soldiers.
Dice	Each side rolls min(attacker army − 1, defender army) dice. Compare corresponding pairs; lower dice lose a soldier, while ties cause no loss.
Capture	Reward = max(2, floor(original country soldier value / 2)). One earned soldier garrisons the captured country.
Transfers	Only between the same commander's countries, outside active combat; retain at least one soldier at the origin.
End conditions	Stopping an attack ends the turn. A commander with no countries is eliminated. Ownership of all ten wins.

Additional features

- Four difficulty levels: Easy, Medium, Hard and Extreme.
- Six-character multiplayer room codes, country reservations and bot-filled seats.
- Three-round NPC alliances with trust, expiry and betrayal.
- Captain Mara's guided tutorial with typed explanations and control pointers.
- Live spectator links, player-left notices and completed-game recordings.
- Replay, flags, distant-attack planes, music and reduced-motion settings.

> Example: an attacker with 10 soldiers against a defender with 2 rolls two pairs. Results 1 vs 2 and 4 vs 3 remove one soldier from each country, leaving 9 vs 1.

Getting started and configuration

Local installation and hosted operation

Prerequisites and installation

Use Node.js 24 with npm, a modern JavaScript-enabled browser and internet access for Firebase services. Java 21 or later is needed only for Firestore emulator tests. The submission ZIP can be extracted instead of cloning.

```
git clone https://github.com/Muhammed112/Meridian_Conquest_UI.git
cd Meridian_Conquest_UI
npm ci
npm run dev
```

Open **http://localhost:8081**. No build step or administrator CLI login is required for ordinary gameplay and player sign-in. The development server binds to loopback and prints its address.

To use port 8082 in PowerShell:

```
$env:MERIDIAN_PORT = '8082'
npm run dev
```

Environment configuration

Location	Purpose
`dist/firebase-config.js`	Public Firebase web configuration for the connected project. It is not an administrator credential.
`dist/ai-config.js`	Enabled flag, configured Gemini model and public reCAPTCHA Enterprise site key.
`dist/ai-local-config.js`	Empty public stub; the development server supplies local App Check configuration in memory.
`.firebaserc`	Firebase project alias; the connected project is meridian-conquest-m112.
`MERIDIAN_PORT`	Optional local server port; defaults to 8081.

The browser app does not require a personal Gemini API key or a new `.env.local` file. Credentials and private App Check tokens must stay outside public assets and the submission package.

Local AI and deployment

An operator with access to the configured Firebase project can run `npx firebase login` and restart `npm run dev` to configure a private local App Check attestation. Reviewers without operator access can use the deployed website for commentary.

Vercel serves `dist` using `vercel.json`, with framework “Other” and empty build/install commands. Firebase Authentication and App Check must allow the deployed hostname. Changes to Firestore permissions are tested and deployed separately from the frontend.

> Firebase remains on Spark. The optional `functions/` implementation is not deployed. Ordinary checks do not publish a site or change production permissions.

System components and database

Responsibilities, encapsulation and persistence

Component	Responsibility and representative API
`GameEngine` `dist/engine.js`	Applies rules and records facts: reinforce(), transfer(), startAttack(), roll(), stopAttack(), endTurn() and proposeAlliance().
`BattleEngine` / `NpcStrategy`	BattleEngine.resolve() compares simultaneous dice. NPC plans evaluate force, rewards, threats, reserves and alliances without changing dice.
`CampaignStore`	list(), open() and save() separate board summaries from paginated event histories and detect competing saves.
Room service	Country choices, opening dice, host-controlled bot turns, room revisions and transactions.
`CommentaryQueue`	enqueue() and drain() manage event priority, deduplication, cached reports and asynchronous generation.
UI and presentation	Shared render helpers/controllers, map interactions, typed tutorial dialogue, MotionDirector and SoundDirector.

Cloud Firestore collections

Path	Purpose / access
`users/{uid}/games/{gameId}`	Private campaign board, listing fields and save revision; signed-in owner access.
`…/events/{eventId}`	Immutable factual event, sequence and predecessor link.
`…/aiReportsV2/{eventId}`	Separate narration with event reference, model, version and server timestamp.
`rooms/{code}`	Members, country reservations, opening dice and shared board; controlled member updates.
`rooms/{code}/events` `rooms/{code}/aiReportsV2`	Room event history and host-generated commentary.
`liveGames/{gameId}` `…/presence/{sessionId}`	Explicit public projection and owner-written presence heartbeats.
`publicData/populations`	Validated public population reference; restricted updater writes.

Save and validation design

Campaign headers load independently from history. Selected histories load in pages of up to 250 events. New events are staged immutably; a transaction commits the new revision and history head. Readers follow the committed chain, excluding unfinished or competing branches. Legacy records remain readable.

Firebase Authentication manages passwords. Firestore rules enforce ownership, membership, turn permissions and bounded transitions. Generated text is excluded from immutable combat facts. Public projections select gameplay fields and exclude account email addresses and credentials.

User roles and workflows

Players, room members, visitors and operators

Role	Permitted interaction
Signed-in player	Create and resume private campaigns, play turns, manage a nickname and explicitly share a campaign.
Multiplayer host	Create/start a room, prepare opening rolls, control bot turns and generate shared room narration.
Multiplayer friend	Join by code, choose a distinct country, roll individually and act only during their own turn.
Public spectator	Read an explicitly shared live or recorded board without sign-in; cannot play or negotiate.
Project operator	Configure Firebase/hosting, run deliberate integration tools and publish tested releases; not an in-game administrator role.

Single-player and tutorial

1. Create an account or sign in and choose a nickname. New players without campaigns are offered guided practice.
2. Select New campaign, a difficulty and a searchable starting country; inspect the nine random rivals.
3. Roll, resolve tied leaders and inspect Turn order. Select Start campaign.
4. From round two, allocate reinforcements. Transfer, attack one target, negotiate an NPC pact or skip as permitted.
5. Read factual reports or AI narration. Resume the campaign later or inspect chronological replay.

Multiplayer and alliances

The host shares a six-character invitation code. Friends sign in and reserve distinct countries. The host prepares opening dice; empty seats become bots. Each human rolls, tied leaders reroll and the host starts after the order is resolved. The host keeps the room open for NPC playback.

Select **Alliance** during a clear planning phase to offer an NPC a three-round pact. One ally is allowed at a time. Armies and turns remain separate. Attacking an ally breaks the pact and reduces trust. Expiry, elimination or only the allies remaining ends the agreement. Human-to-human proposals are outside this version.

Sharing, reports and viewing

Share live link publishes an ongoing single-player board. **Stop sharing** removes its public copy without deleting the private campaign. **Share gameplay** creates a fixed completed-game recording with up to 60 recent events and saved commentary; that embedded copy cannot be revoked.

Pin reports keeps a bounded feed beneath a proportionally scaled map. Both remain in page flow and move together when the page scrolls; messages scroll internally. A missing owner heartbeat produces a player-left notice, with abrupt exits detected within about 80 seconds.

Verification and troubleshooting

Reproducible checks and practical boundaries

Command	Purpose
`npm run check`	Formatting, ESLint and unit tests.
`npm run browsers:install` `npm run test:browser`	Install Chromium, then run isolated browser gameplay/UI fixtures. Installed Chrome is also supported.
`npm run test:rules`	Firestore permission tests against demo-meridian on port 8180; requires Java 21+.
`npm run test:ai`	Offline commentary-validation cases, without a model request.
`npm run commentary:preview`	Deliberate real-Gemini evaluation; needs operator/local setup and uses provider quota.

The latest recorded release checks passed 273 unit tests and 18 permission tests. Published-file verification matched 304 non-audio assets, checked seven private paths, and exercised the homepage and opening preview. Passing checks cover their fixtures; they do not prove every possible interaction correct.

Troubleshooting

Symptom	Check
Old labels or layout	Refresh with Ctrl+F5 after an update.
Local port unavailable	Set MERIDIAN_PORT to an unused port and restart the server.
Local AI unavailable	Use npm run dev with operator App Check setup, or try the deployed site. Check the visible AI status and provider quota.
Campaigns missing	Confirm the signed-in account; private saves are account-specific.
Friend cannot open localhost link	Share from the deployed site. Loopback addresses refer to the viewer’s own computer.
Multiplayer bots stop	Keep the host’s room open and connected.

Known limitations and final-review follow-up

- Dice and full combat execute in the browser. Rules enforce substantial storage/turn constraints, but there is no authoritative competitive anti-cheat server.
- Gemini response time and quotas vary. Event priorities skip stale narration; factual reports and gameplay continue independently.
- Population refresh failures retain the last validated snapshot. Existing games keep their saved populations and rewards.
- The final source audit reproduced a delayed sharing-restoration race that can reinstate a revoked share. A guard and regression case remain to be added.
- Other audit follow-ups concern multiplayer viewer-specific sound cues, legacy narration cache precedence, operator argument validation and live-test fixture maintenance. These findings were not fixed as part of preparing this document.

Project structure and references

Source organisation, supporting guides and credits

```
Meridian_Conquest_UI/
  dist/                  Browser source and bundled assets
    controllers/         Actions, campaign loading, NPC playback
    ui/                  Shared screens, map, reports and tutorial
  scripts/               Dev server, data tools, browser checks
  tests/                 Unit and Firestore permission tests
  docs/                  Architecture, setup and user guidance
  functions/             Optional undeployed commentary backend
  firestore.rules        Firestore access and transition rules
  firestore.indexes.json Index settings
  firebase.json          Firebase deployment/emulator settings
  vercel.json            Static hosting configuration
  package.json           Dependencies and project commands
  README.md              Installation and usage guide
  CONTRIBUTING.md        Development practices
```

Additional documentation

File	Purpose
`README.md` / `CONTRIBUTING.md`	Installation, usage, support and development checks.
`docs/ARCHITECTURE.md` `docs/DATABASE.md`	Responsibilities, class boundaries, schema, save compatibility and access.
`docs/FIREBASE.md` `docs/VERCEL.md`	Service setup, hostname registration and deployment procedures. Historical release sections are dated.
`docs/TUTORIAL.md` `docs/DIPLOMACY.md`	Instruction coverage and pact behaviour. Older guides may call the Alliance panel “Diplomacy”.
`docs/email-templates/`	Password-reset template and its setup guidance.

Author, support and contributions

Developed by **Muhammed (Muhammed112)** for the INF4001N entrance assignment. Report reproducible issues through the repository’s issue tracker. Contributions should preserve game rules and save compatibility, reuse shared logic and include appropriate checks.

Acknowledgements and licences

Population figures: Worldometers. Country metadata: mledoze/countries (ODbL 1.0). Map: world-atlas (ISC) and Natural Earth geometry. Flags: flag-icons (MIT). Music: Kevin MacLeod’s Decisions, Lost Frontier and Mechanolith (CC BY 4.0). Sound effects: Kenney CC0 samples and original Web Audio cues. The interface builds on the supplied Figma design.

Asset notices remain in `dist/assets`; music attribution is also visible in Sound settings. No separate project-wide open-source licence is included for the original application code.

Package source, lockfiles, configuration, tests and asset notices. Exclude private folders, dependencies, environment files and logs. Supply test credentials and examiner access separately through the university.